10-3. CATE: Heterogeneous Treatment Effects

A consolidated, mathematical lecture companion

English notes based on the lecture slides *CATE: Heterogeneous Treatment Effects* (Prof. Sanghack Lee, GSDS SNU)

The central question is how treatment effects vary with covariates, why that matters, and how flexible machine-learning "meta-learners" estimate the conditional average treatment effect $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$.

These notes consolidate the slides, add the arithmetic of the motivating example, and state each meta-learner (T, S, regression imputation, X, DR) with the reasoning behind it. Notation: X treatment, Z covariates, $e(z) = P(X = 1 \mid Z = z)$. Physical PDF page numbers are listed in Section 3.

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1 Why heterogeneous treatment effects?

Purpose. Show that a single average can hide opposite effects in subgroups.

1.1 The average can be misleading

Consider 20 individuals, half women ($Z = 1$) and half men ($Z = 0$), with a null average effect:

$$P(Y(1) = 1) - P(Y(0) = 1) = 0.5 - 0.5 = 0.$$

Yet among women the causal risk difference is $0.6 - 0.4 = +0.2$ (treatment increases the risk of death) and among men $0.4 - 0.6 = -0.2$ (it decreases it). Indeed $\frac{1}{2}(+0.2) + \frac{1}{2}(-0.2) = 0$. *There is generally no such thing as "the causal effect of treatment"; it depends on the characteristics of the population studied* (Hernán and Robins, *What If*, §4.1).

1.2 Effect modification

Definition 1.1. Z is a *modifier* of the effect of X on Y when the average causal effect of X on Y varies across levels of Z.

- **Additive** effect modification: $\mathbb{E}[Y(1) - Y(0) \mid Z = 1] \neq \mathbb{E}[Y(1) - Y(0) \mid Z = 0]$.
- **Multiplicative** effect modification: $\dfrac{\mathbb{E}[Y(1) \mid Z = 1]}{\mathbb{E}[Y(0) \mid Z = 1]} \neq \dfrac{\mathbb{E}[Y(1) \mid Z = 0]}{\mathbb{E}[Y(0) \mid Z = 0]}$.

Modification may appear on one scale but not on the other (e.g. risks $0.2 \to 0.4$ and $0.4 \to 0.8$ have equal ratios but different differences). *Qualitative* effect modification (effects in opposite directions across strata) implies modification on both scales.

1.3 Why care?

1. **Policy targeting:** identify who benefits. If treatment harms women and helps men (qualitative modification), a physician would, absent other information, treat only men.
2. **Transportability:** if the distribution of modifiers (sex, age) differs between the study and a target population, so does the ATE.
3. **Mechanisms:** modification may reveal biological, social or other mechanisms.

Conditioning on the full covariate vector gives the CATE, $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$ (*What If*, §4.3).

1.4 From stratification to CATE estimation

The traditional approach computes the effect within each stratum of Z and compares stratum-specific measures. Its limitations: continuous modifiers must be discretised; the number of strata grows exponentially with the number of covariates (curse of dimensionality); strata may contain too few observations. Modern CATE estimators (meta-learners, causal forests) use flexible ML models to estimate $\tau(z)$ without explicit stratification (*What If*, §4.2).

2 Meta-learners for CATE

Purpose. Build CATE estimators from arbitrary regression algorithms (Künzel et al. 2019, PNAS).

Under unconfoundedness and positivity, $\mu_x(z) := \mathbb{E}[Y(x) \mid Z = z] = \mathbb{E}[Y \mid X = x, Z = z]$, so any regression method can be used.

2.1 T-learner and S-learner

T-learner (two models): fit $\hat{\mu}_1(z)$ on the treated and $\hat{\mu}_0(z)$ on the controls separately;

$$\hat{\tau}_T(z) = \hat{\mu}_1(z) - \hat{\mu}_0(z), \qquad \widehat{\mathrm{ATE}}_T = \frac{1}{n}\sum_{i=1}^{n}\big(\hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i)\big).$$

S-learner (single model): fit one model $\hat{\mu}(z, x) \approx \mathbb{E}[Y \mid Z = z, X = x]$ with X as an ordinary feature;

$$\hat{\tau}_S(z) = \hat{\mu}(z, 1) - \hat{\mu}(z, 0), \qquad \widehat{\mathrm{ATE}}_S = \frac{1}{n}\sum_{i=1}^{n}\big(\hat{\mu}(Z_i, 1) - \hat{\mu}(Z_i, 0)\big).$$

The T-learner can be noisy when one arm is small; the S-learner may shrink the effect of X towards zero if the learner regularises it away.

2.2 Regression imputation

Plug in $\hat{\mu}$ only for the *unobserved* counterfactual and use Y_i for the observed one:

$$\hat{\tau}_{RI} = \frac{1}{n}\sum_{i=1}^{n}\Big[\big(X_i Y_i + (1 - X_i)\hat{\mu}_1(Z_i)\big) - \big((1 - X_i)Y_i + X_i\hat{\mu}_0(Z_i)\big)\Big].$$

For treated units ($X_i = 1$): observed Y_i for $Y(1)$, imputed $\hat{\mu}_0(Z_i)$ for $Y(0)$; for controls: imputed $\hat{\mu}_1(Z_i)$ for $Y(1)$, observed Y_i for $Y(0)$.

2.3 X-learner

1. Estimate the response functions $\hat{\mu}_0(z) \approx \mathbb{E}[Y(0) \mid Z = z]$ and $\hat{\mu}_1(z) \approx \mathbb{E}[Y(1) \mid Z = z]$.
2. Impute individual treatment effects: for the treated $\tilde{D}_i^1 := Y_i - \hat{\mu}_0(z_i)$; for the controls $\tilde{D}_i^0 := \hat{\mu}_1(z_i) - Y_i$. Regress to obtain $\hat{\tau}_1(z) \approx \mathbb{E}[\tilde{D}_i^1 \mid Z = z]$ (from the treated) and $\hat{\tau}_0(z) \approx \mathbb{E}[\tilde{D}_i^0 \mid Z = z]$ (from the controls).
3. Combine with a weight function $g(z) \in [0, 1]$, e.g. $g(z) = \hat{e}(z)$:

$$\hat{\tau}(z) = g(z)\,\hat{\tau}_0(z) + (1 - g(z))\,\hat{\tau}_1(z).$$

Illustration. In the slides' simulation (one covariate $x \in [-1, 1]$) the true effect is $\tau = 1$ and treated units are rare ($g = \hat{e} \approx 0.05$). Step 1 fits $\hat{\mu}_0$ as a piecewise-linear function (many control data) and $\hat{\mu}_1$ as a linear function (few treated data). $\hat{\mu}_0$, fitted on the large control group, is flexible and contains jumps; $\hat{\mu}_1$, fitted on few treated points, is nearly linear. The T-learner $\hat{\tau}_T = \hat{\mu}_1 - \hat{\mu}_0$ inherits the jumps of $\hat{\mu}_0$, while the X-learner $\hat{\tau}_X \approx (1 - g)\hat{\tau}_1$ is nearly linear and close to the true τ: $\hat{\tau}_1$ uses the imputed effects of the treated, computed against the accurate $\hat{\mu}_0$, and is smooth because the effect itself is simple.

2.4 DR-learner

Apply the doubly robust score of Lecture 10-2 to learn $\tau(z)$:

1. Estimate the nuisances $\hat{\mu}_x(z)$ and $\hat{e}(z)$.
2. Construct the pseudo-outcome
$$\tilde{Y}_i = \frac{X_i(Y_i - \hat{\mu}_1(Z_i))}{\hat{e}(Z_i)} - \frac{(1 - X_i)(Y_i - \hat{\mu}_0(Z_i))}{1 - \hat{e}(Z_i)} + \hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i).$$
3. Regress $\tilde{Y}_i$ on Z_i with any flexible method to obtain $\hat{\tau}(z)$.

Why the DR-learner?

- $\mathbb{E}[\tilde{Y}_i \mid Z_i = z] = \tau(z)$ when the nuisances are correct (by the same computation as for the DR estimator, conditionally on Z); in fact one correct nuisance suffices.
- *Neyman orthogonality:* errors in $\hat{\mu}, \hat{e}$ affect $\hat{\tau}$ only at second order (through products of the two errors).
- It adapts to the complexity of $\tau(z)$, even when the nuisance functions are more complex (Kennedy 2023).
- In practice use K-fold cross-fitting: estimate the nuisances on folds $\neq k$ and construct $\tilde{Y}_i$ on fold k, to avoid overfitting bias.

3 Source coverage

The source deck has 29 physical pages (10 numbered slides with build-ups).

Topic	Physical pages
Misleading averages, effect modification, motivation, stratification	1–15
T/S-learners, regression imputation	16–20
X-learner and illustration	21–26
DR-learner	27–29